

SYSTEMS ANALYSIS & DESIGN

Information Systems

What an information system is, why organizations need one, a detailed look at all seven major types with examples, and how they compare.

What Is an Information System?

An Information System (IS) is a structured combination of hardware, software, data, people, and processes that work together to collect, store, process, and distribute information. It is designed to support decision-making, coordination, analysis, and control within an organization.

Everyday Examples

1

Banking System

Processes customer transactions such as deposits, withdrawals, and fund transfers.

2

Inventory Management

Tracks product stock levels to keep operations running smoothly.

3

Healthcare System

Manages patient records to ensure accurate, timely care.

Why Is an Information System Necessary?

1

Saves Time

Automates routine tasks such as tracking sales, managing employees, and storing customer details.

2

Improves Decision-Making

Provides accurate, updated information so managers can analyze performance and plan ahead.

3

Enables Better Communication

Lets employees share data easily, work together, and serve customers more efficiently.

4

Keeps Data Safe

Prevents unauthorized access and cyber threats, protecting the organization's information.

Overall, an information system makes work easier, faster, and more secure.

Transaction Processing System (TPS)

A Transaction Processing System is designed to handle and record routine, day-to-day business transactions. It ensures the smooth processing of operations like sales, payroll, and order processing, and forms the foundation of many other business processes.

Key Characteristics

Reliable

Fast

Accurate

High Volume

Foundation for MIS

Real-World Examples

Banking System

Processes withdrawals, deposits, and fund transfers for millions of customer accounts every day.

Retail POS System

Records every sales transaction in a supermarket at the checkout counter in real time.

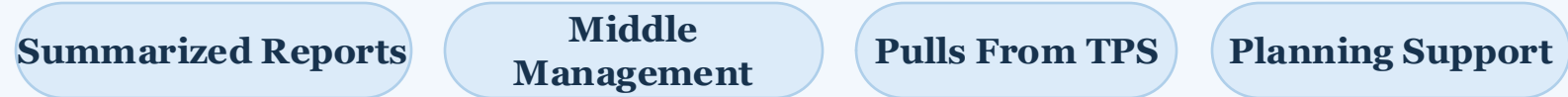
Online Booking System

Manages airline or hotel reservations, seat/room availability, and payment confirmation.

Management Information System (MIS)

A Management Information System collects and processes data from TPS and other sources to generate reports that help managers monitor business performance. It provides summarized, structured information for middle management to support planning and decision-making.

Key Characteristics



Real-World Examples

Sales Reports

Summarizes monthly revenue figures and customer purchase trends for review meetings.

Employee Performance Reports

Tracks staff attendance and productivity to support appraisals and scheduling.

Inventory Management

Monitors stock levels across locations to prevent shortages or overstocking.

Decision Support System (DSS)

A Decision Support System helps managers and executives make informed decisions by analyzing large amounts of data and providing insights, forecasts, and recommendations. It often includes data visualization tools, simulations, and artificial intelligence techniques.

Key Characteristics

Model-Driven

Data-Driven

Document-Driven

Knowledge-Driven

Executive-Level

Real-World Examples

Stock Market Analysis

Predicts stock price trends based on historical market data for traders and analysts.

Medical Diagnosis System

Assists doctors in identifying possible diseases from patient symptoms and history.

Supply Chain Optimization

Helps companies choose the best suppliers and transportation routes for cost and speed.

Enterprise Resource Planning (ERP)

An Enterprise Resource Planning System integrates different business functions — finance, HR, sales, and supply chain — into a single system. It allows departments to share data in real time, improving overall efficiency and coordination across the organization.

Key Characteristics

Cross-Department

Real-Time Data

Single System

Improves Coordination

Real-World Examples

SAP ERP System

Used by large companies worldwide to manage core business operations end-to-end.

Oracle ERP

Manages finance, manufacturing, and customer relations within one integrated platform.

University ERP

Handles student admissions, payroll processing, and library management on campus.

Customer Relationship Management (CRM)

A Customer Relationship Management System is designed to manage customer interactions, track sales, and improve customer service. Businesses use CRM systems to maintain strong relationships with clients and increase overall customer satisfaction.

Key Characteristics

Customer-Centric

Sales Tracking

Lead Management

Service Quality

Real-World Examples

Salesforce CRM

Manages customer interactions and tracks sales leads through the entire pipeline.

HubSpot CRM

Organizes customer data and coordinates inbound marketing campaigns in one place.

Call Center CRM

Records customer service requests and responses for consistent follow-up.

Supply Chain Management (SCM)

A Supply Chain Management System helps businesses manage the flow of goods, services, and information across the supply chain. It ensures that products are delivered efficiently from suppliers all the way through to customers.

Key Characteristics

Logistics

Supplier Coordination

Inventory Flow

Delivery Efficiency

Real-World Examples

Amazon's SCM System

Manages warehouses and coordinates deliveries across a vast distribution network.

Walmart's Inventory System

Automatically restocks products across stores based on real-time demand data.

Logistics Tracking System

Monitors shipments in real time from origin to final delivery destination.

Expert System (ES)

An Expert System is designed to mimic human decision-making by using artificial intelligence and knowledge-based rules. It helps solve complex problems in specialized fields such as medicine, engineering, and finance, guided by an expert-built knowledge base.

Key Characteristics

AI-Powered

Knowledge Base

Specialized Fields

Mimics Experts

Real-World Examples

Medical Expert System

Diagnoses diseases by matching patient symptoms against a medical knowledge base.

Financial Fraud Detection

Identifies suspicious transactions in real time using rule-based pattern analysis.

Engineering Design System

Assists engineers in complex machine designs by applying expert design rules.

How the Seven Systems Compare

System	Primary Purpose	Main Users	Data Source	Example
TPS	Record routine daily transactions	Operational staff	Direct transaction entry	Bank withdrawals/deposits
MIS	Summarize data for monitoring	Middle management	TPS & internal records	Monthly sales reports
DSS	Support complex decisions	Managers & executives	Internal + external + models	Stock market forecasting
ERP	Integrate business functions	All departments	Shared enterprise database	SAP, Oracle, University ERP
CRM	Manage customer relationships	Sales & service teams	Customer interaction data	Salesforce, HubSpot
SCM	Coordinate the supply chain	Logistics & operations	Supplier & shipment data	Amazon SCM, Walmart
ES	Mimic expert decision-making	Specialists (medicine, finance)	Expert-built knowledge base	Medical diagnosis systems

KEY TAKEAWAYS

Information Systems at a Glance

- 1 An Information System combines hardware, software, data, people, and processes to collect, store, process, and distribute information.
- 2 It exists to save time, improve decisions, enable communication, and keep data secure.
- 3 TPS, MIS, and DSS form a hierarchy — from recording daily transactions to summarizing them to supporting executive decisions.
- 4 ERP, CRM, and SCM each coordinate a different slice of the organization: internal operations, customers, and suppliers.
- 5 Expert Systems apply AI and codified expert knowledge to solve specialized problems that once required a human specialist.